```
!pip install scikit-survival lifelines matplotlib seaborn pandas numpy
```

```
Collecting scikit-survival
  Downloading scikit_survival-0.27.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (7.0 kB)
Collecting lifelines
  Downloading lifelines-0.30.3-py3-none-any.whl.metadata (3.5 kB)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Collecting ecos (from scikit-survival)
  Downloading ecos-2.0.14-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (8.0 kB)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scikit-survival) (1.5.3)
Requirement already satisfied: numexpr in /usr/local/lib/python3.12/dist-packages (from scikit-survival) (2.14.1)
Requirement already satisfied: osqp>=1.0.2 in /usr/local/lib/python3.12/dist-packages (from scikit-survival) (1.1.1)
Requirement already satisfied: scipy>=1.13.0 in /usr/local/lib/python3.12/dist-packages (from scikit-survival) (1.16.3)
Collecting scikit-learn<1.9,>=1.8.0 (from scikit-survival)
  Downloading scikit_learn-1.8.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Requirement already satisfied: autograd>=1.5 in /usr/local/lib/python3.12/dist-packages (from lifelines) (1.8.0)
Collecting autograd-gamma>=0.3 (from lifelines)
  Downloading autograd-gamma-0.5.0.tar.gz (4.0 kB)
  Preparing metadata (setup.py) ... done
Collecting formulaic>=0.2.2 (from lifelines)
  Downloading formulaic-1.2.1-py3-none-any.whl.metadata (7.0 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
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Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
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Collecting interface-meta>=1.2.0 (from formulaic>=0.2.2->lifelines)
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Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from osqp>=1.0.2->scikit-survival) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from osqp>=1.0.2->scikit-survival) (75.2
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Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn<1.9,>=1.8.0-
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Downloading ecos-2.0.14-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (222 kB)
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Downloading interface_meta-1.3.0-py3-none-any.whl (14 kB)
Building wheels for collected packages: autograd-gamma
  Building wheel for autograd-gamma (setup.py) ... done
  Created wheel for autograd-gamma: filename=autograd_gamma-0.5.0-py3-none-any.whl size=4030 sha256=ac79efc3d3f20812011cc7ffa7
  Stored in directory: /root/.cache/pip/wheels/50/37/21/0a719b9d89c635e89ff24bd93b862882ad675279552013b2fb
```
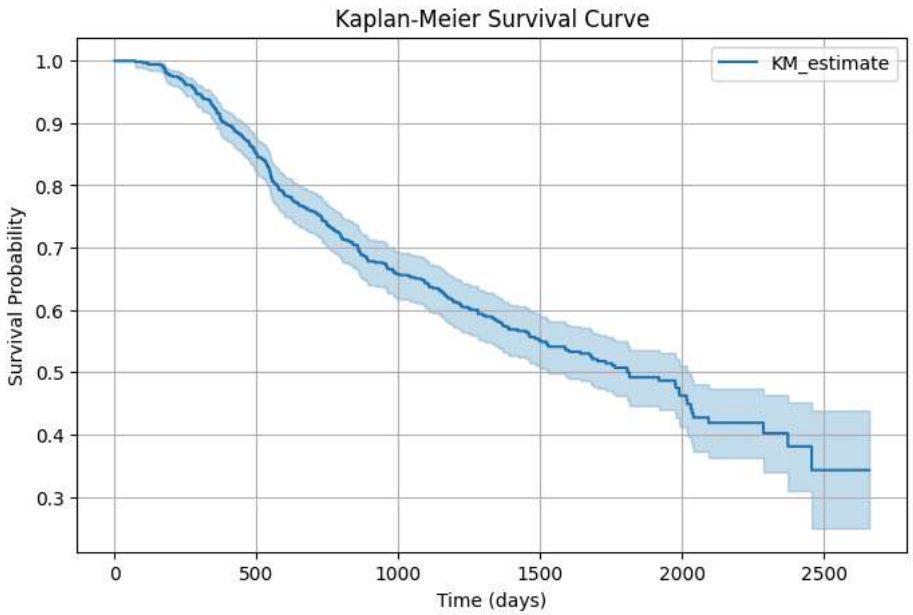

```
from sksurv.datasets import load_gbsg2
import pandas as pd

X, y = load_gbsg2()
print(X.head())
print(y[:5])  # Shows (event_occurred, survival_time) tuples
```

```
    age  estrec horTh menostat  pnodes  progrec tgrade  tsize
0  70.0    66.0    no     Post     3.0     48.0     II   21.0
1  56.0    77.0   yes     Post     7.0     61.0     II   12.0
2  58.0   271.0   yes     Post     9.0     52.0     II   35.0
3  59.0    29.0   yes     Post     4.0     60.0     II   17.0
4  73.0    65.0    no     Post     1.0     26.0     II   35.0
[( True, 1814.) ( True, 2018.) ( True,  712.) ( True, 1807.)
 ( True,  772.)]
```

```
from lifelines import KaplanMeierFitter
import matplotlib.pyplot as plt
```

```
# Extract time and event status
times = [t for (e, t) in y]
events = [e for (e, t) in y]

kmf = KaplanMeierFitter()
kmf.fit(times, event_observed=events)

plt.figure(figsize=(8, 5))
kmf.plot_survival_function()
plt.title("Kaplan-Meier Survival Curve")
plt.xlabel("Time (days)")
plt.ylabel("Survival Probability")
plt.grid(True)
plt.show()
```




```
from sklearn.preprocessing import OrdinalEncoder
from sklearn.model_selection import train_test_split
import numpy as np

# Encode categorical columns
X_encoded = X.copy()
cat_cols = X.select_dtypes(include='category').columns
enc = OrdinalEncoder()
X_encoded[cat_cols] = enc.fit_transform(X[cat_cols])

X_train, X_test, y_train, y_test = train_test_split(
    X_encoded, y, test_size=0.2, random_state=42
)
```

```
from sksurv.linear_model import CoxPHSurvivalAnalysis

cox_model = CoxPHSurvivalAnalysis(alpha=0.01)
cox_model.fit(X_train, y_train)

cox_cindex = cox_model.score(X_test, y_test)
print(f"Cox PH C-index: {cox_cindex:.4f}")
```

```
Cox PH C-index: 0.6493
```

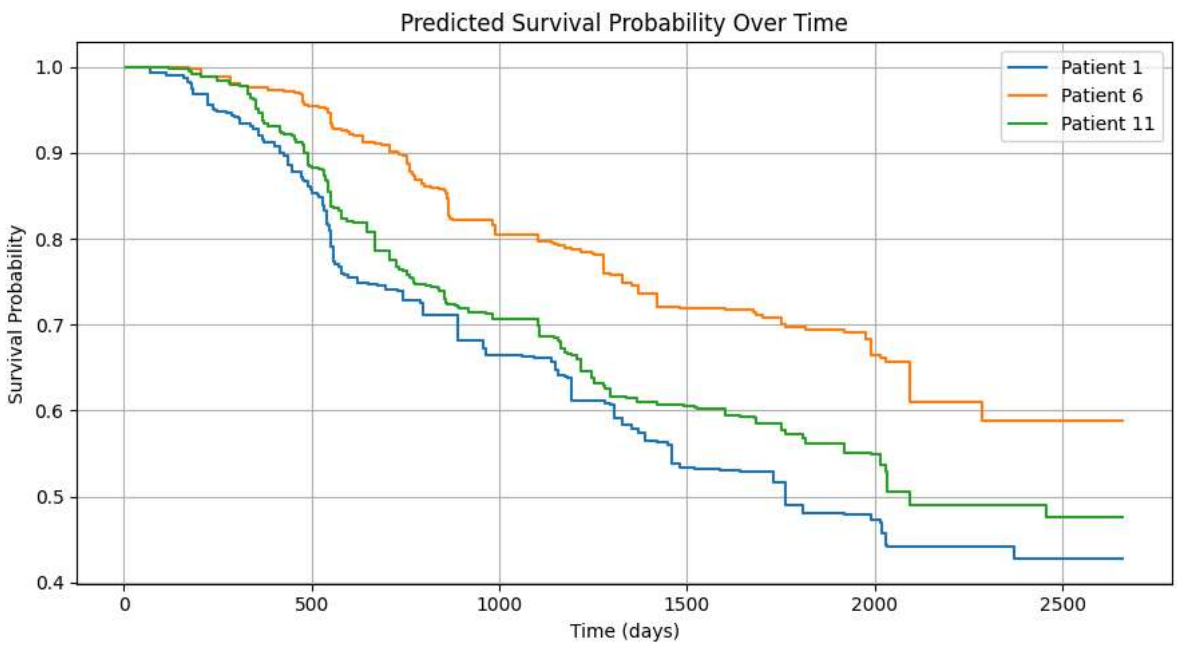
```
from sksurv.ensemble import RandomSurvivalForest

rsf = RandomSurvivalForest(n_estimators=100, min_samples_split=10,
                           min_samples_leaf=15, random_state=42, n_jobs=-1)
rsf.fit(X_train, y_train)

rsf_cindex = rsf.score(X_test, y_test)
print(f"Random Survival Forest C-index: {rsf_cindex:.4f}")
```

```
Random Survival Forest C-index: 0.6624
```

```
print(f"\nModel Comparison:")
print(f"Cox PH Model     C-index: {cox_cindex:.4f}")
print(f"Survival Forest  C-index: {rsf_cindex:.4f}")

# Higher C-index = better (0.5 = random, 1.0 = perfect)
```

```
Model Comparison:
Cox PH Model     C-index: 0.6493
Survival Forest  C-index: 0.6624
```

```
# Pick 3 test patients and plot their survival probability curves
sample_idx = [0, 5, 10]
times_to_pred = np.arange(1, 2000, 50)

plt.figure(figsize=(9, 5))
for i in sample_idx:
    surv_fn = rsf.predict_survival_function(X_test.iloc[[i]])[0]
    plt.step(surv_fn.x, surv_fn(surv_fn.x), where="post", label=f"Patient {i+1}")

plt.xlabel("Time (days)")
plt.ylabel("Survival Probability")
plt.title("Predicted Survival Probability Over Time")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Start coding or generate with AI.